| Question | | | | Marking details | Marks Available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **3** | (a) | | | **Indicative content:**<br><br>Collect some water in a large container – about 2-3cm deep.<br>Collect samples of invertebrates using the net and transfer them to the tray.<br>Study the organisms in the tray and try and identify the invertebrates against the chart.<br>Record the number of each invertebrate that has been caught.<br>Compare with the key to determine water quality according to numbers of indicator species present.<br>Pour them and the water gently back into the stream.<br><br>**5– 6 marks**<br>Comprehensive account of the method of collection and determination of water quality and reference to conclusions.<br><br>*There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.*<br><br>**3 – 4 marks**<br><br>Full method of collection and how to determine water quality OR most of method and conclusion.<br><br>*There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.*<br><br>**1-2 marks**<br>Limited account of collection method or how to determine water quality.<br><br>*There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.*<br><br>**0 marks**<br>No attempt made or no response worthy of credit. | 6 | | | 6 | | 6 |

| Question | | | | Marking details | AO1 | AO2 | AO3 | Total | Maths | Prac |
|---|---|---|---|---|---|---|---|---|---|---|
| | (b) | (i) | | Stream A has <u>some pollution</u> (1)<br>since majority of invertebrates are freshwater shrimps and caddis fly lava (1).<br>Stream B has <u>highly pollution</u> (1)<br>because the largest population are sludge worms. (1) | | 4 | | 4 | | 4 |
| | | (ii) | | Even if the same group of students (used the same method) (1)<br>results may differ because the ecosystem is not static(1) | | | 2 | 2 | | 2 |
| | | | | **Question 3 total** | **6** | **4** | **2** | **12** | **0** | **12** |